

HDOC - Day 10 Assignment

Question 1: Convert Time in Seconds into Days, Hours, Minutes, and Seconds

1. Algorithm

1. Start.
2. Declare variables: total_seconds, days, hours, minutes, and seconds.
3. Input the value of total_seconds from the user.
4. Calculate $\text{days} = \text{total_seconds} / 86400$.
5. Update $\text{total_seconds} = \text{total_seconds} \% 86400$.
6. Calculate $\text{hours} = \text{total_seconds} / 3600$.
7. Update $\text{total_seconds} = \text{total_seconds} \% 3600$.
8. Calculate $\text{minutes} = \text{total_seconds} / 60$.
9. Calculate remaining seconds = $\text{total_seconds} \% 60$.
10. Print days, hours, minutes, and seconds.
11. Stop.

2. Pseudocode

BEGIN

INTEGER total_seconds, days, hours, minutes, seconds

PRINT "Enter time in seconds: "

READ total_seconds

$\text{days} = \text{total_seconds} / 86400$

$\text{total_seconds} = \text{total_seconds} \text{ MOD } 86400$

$\text{hours} = \text{total_seconds} / 3600$

$\text{total_seconds} = \text{total_seconds} \text{ MOD } 3600$

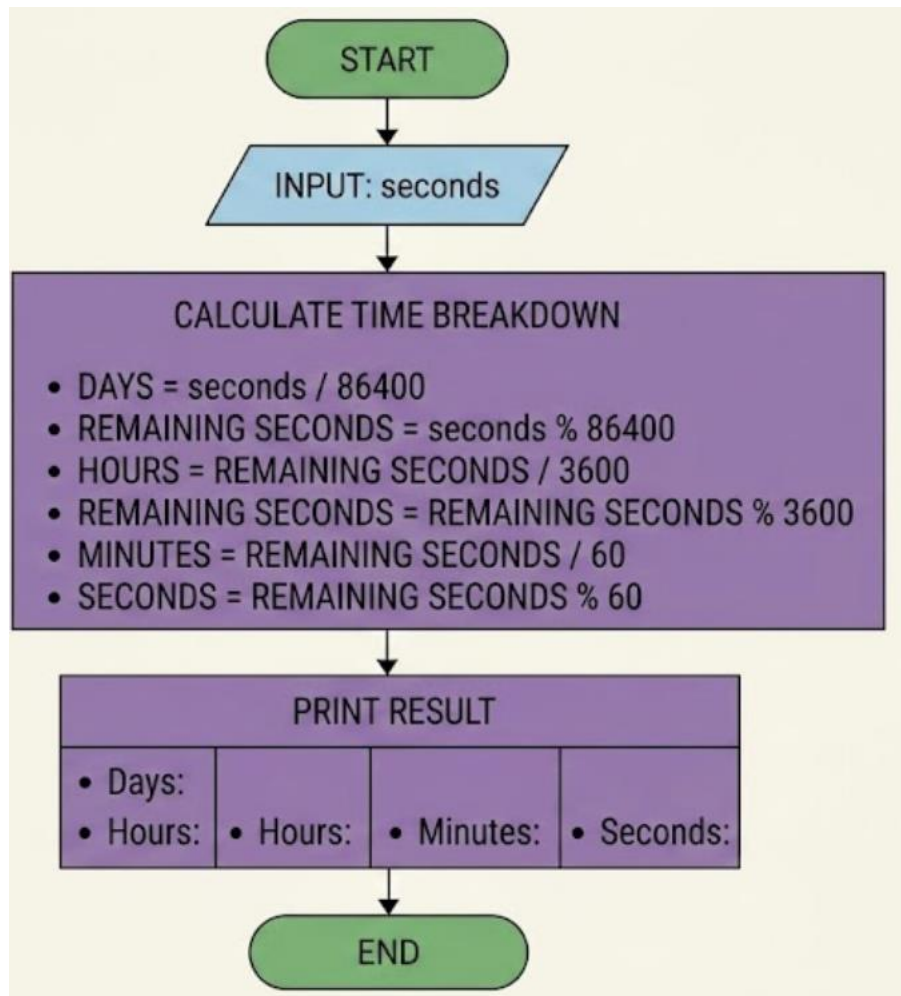
$\text{minutes} = \text{total_seconds} / 60$

$\text{seconds} = \text{total_seconds} \text{ MOD } 60$

PRINT days, hours, minutes, seconds

END

3. Flowchart



4. Test Case Table

Test case	Input	Expected op	Actual op	status
1.	90061	1 Days, 1 Hours, 1 Minutes, 1 Seconds	1 Days, 1 Hours, 1 Minutes, 1 Seconds	pass
2.	3600	0 Days, 1 Hours, 0 Minutes, 0 Seconds	0 Days, 1 Hours, 0 Minutes, 0 Seconds	pass

5. C Code

```
UW PICO 5.09 File: 10q1.c

#include <stdio.h>

int main() {
    long total_seconds, temp_seconds;
    int days, hours, minutes, seconds;

    printf("Enter time in seconds: ");
    scanf("%ld", &total_seconds);

    temp_seconds = total_seconds;

    days = temp_seconds / 86400;
    temp_seconds = temp_seconds % 86400;

    hours = temp_seconds / 3600;
    temp_seconds = temp_seconds % 3600;

    minutes = temp_seconds / 60;
    seconds = temp_seconds % 60;

    printf("\n--- Result ---\n");
    printf("%ld seconds = %d Days, %d Hours, %d Minutes, %d Seconds\n", total_seconds, days, hours, minutes, seconds);

    return 0;
}
```

- EXECUTION AND OUTPUT :

```
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % gcc 10q1.c -o 10q1
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./10q1
Enter time in seconds: 90061

--- Result ---
90061 seconds = 1 Days, 1 Hours, 1 Minutes, 1 Seconds
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ %
```

Question 2: Check Whether a Given Year is a Leap Year or Not

1. Algorithm

1. Start.
2. Declare variable year.
3. Input year from user.
4. Check if year is divisible by 400.
 - a. If yes, it is a leap year.
5. If not divisible by 400, check if year is divisible by 100.
 - a. If yes, it is not a leap year.
6. If not divisible by 100, check if year is divisible by 4.
 - a. If yes, it is a leap year.
 - b. If no, it is not a leap year.
7. Display the result.
8. Stop.

2. Pseudocode

```
BEGIN
  INTEGER year

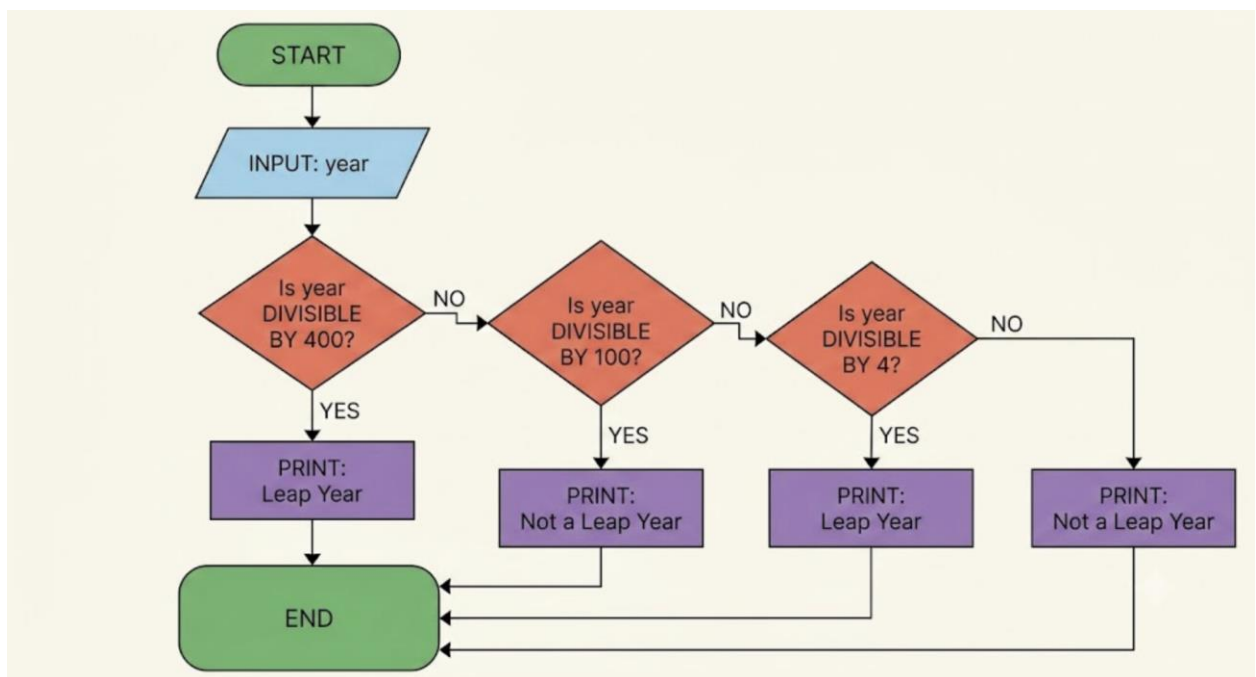
  PRINT "Enter a year: "
  READ year
```

```

IF (year MOD 400 == 0) THEN
    PRINT "Leap Year"
ELSE IF (year MOD 100 == 0) THEN
    PRINT "Not a Leap Year"
ELSE IF (year MOD 4 == 0) THEN
    PRINT "Leap Year"
ELSE
    PRINT "Not a Leap Year"
END IF
END

```

3. Flowchart



4. Test Case Table

TEST CASE	input	Expected output	Actual output	status
1.	2026	Not a leap year	2026 is not a leap year.	pass

2.	2024	Leap year	2024 is a leap year.	pass

5. C Code

```

UW PICU 5.07

#include <stdio.h>

int main() {
    int year;

    printf("Enter a year: ");
    scanf("%d", &year);

    if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
        printf("%d is a Leap Year.\n", year);
    } else {
        printf("%d is NOT a Leap Year.\n", year);
    }

    return 0;
}

```

- EXECUTION AND OUTPUT ;

```
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % nano 10q2.c  
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % gcc 10q2.c -o 10q2  
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./10q2  
Enter a year: 2026  
2026 is NOT a Leap Year.  
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./10q2  
Enter a year: 2024  
2024 is a Leap Year.  
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % █
```